Mukesh Kumar

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Principal Full-Stack Engineer | Fintech Solutions Architect

Professional Summary

- Software Engineer with over 20+ years of experience in developing scalable, high-performance web and enterprise-level solutions using modern frameworks and cloud-native technologies. Well-versed in building full-stack applications utilizing Java based back-end frameworks and interactive front-end libraries.

- Hands-on expertise deploying enterprise scale solutions across distributed environments and multiple locations to provide highly scalable, low-latency, highly available fault tolerant solutions.

- Knowledge of relational and NoSQL databases including Oracle, Sybase, PostgreSQL, MySQL, MongoDB and SQL Server to support complex data operations. Experienced in modern DevOps practices, integrating CI/CD workflows using Git, Jenkins, Maven etc.

- Extensive knowledge and experience with containerization tools such as Docker and Kubernetes for reliable deployments. Proficiency in Agile and SDLC methodologies, with strong collaboration, debugging, and problem-solving skills in cross-functional development teams.

- Training, experience and understanding of AI tools and technologies. Comfortable with various tools that augment design, development, testing as well as debugging. Also, familiar with Agentic AI tools and techniques such as – Agentic AI Development, AI Architecture Design, Agentic RAG Implementation, Multi-agent Systems, AI observability and Ops, Build No/Low code AI Agents

Skills & Competencies

AI Tools and technologies: Claude code, Antigravity, VS Code + Copilot, Cursor, V0, Bolt, Manus, Devin and Viktor
Agentic AI Tools: LangChain/LangGraph/LangSmith, LanhFuse, LangFlow, LangGraph, Amazon Bedrock, CrewAI

Frameworks/Libraries: Spring Boot 3+, Spring Framework, Spring Boot, Spring Data/JPA, Hibernate, Spring MVC, Microservices, Angular, Node.js, React.js, Express.js, REST APIs, J2EE technologies

Methodologies: Agile/Scrum, Waterfall

Programming Languages: Java 11+, C++, SQL, Linux scripting languages

Middleware, Messaging and Caching Technologies: Kafka, Tibco EMS, ECS, Coherence, Redis and Gemfire

Web Technologies: HTML5, CSS3, JavaScript, Tailwind, TypeScript, JSON, jQuery, Ajax, XML, MERN and MEAN stack

Application Server: Apache Tomcat, IBM Web Sphere, WebLogic

Databases: Oracle, Sybase, MySQL, SQL Server, MongoDB, PostgreSQL

Version Control: Git, GitHub, GitLab

DevOps and Cloud: Maven, Jira, Jenkins, Junit, Excel, JMeter, Docker, Kubernetes, AWS, Native Cloud

Operating Systems: Windows, Linux, Unix

Career History

Synaptics – Equity derivatives – pricing and back-testing platform

Sept 2023 – current
Self-employed

Orchestrated the end-to-end architectural design and development of a high-performance equity derivatives pricing and back-testing platform from the ground up.
Architected a sophisticated suite of financial services providing real-time price discovery, correlation matrices, and advanced volatility modeling, including skew/term structures and forward factor time-series.
Acted as the single-threaded owner for the product roadmap, translating complex quantitative finance requirements into a scalable, cloud-native microservices architecture.
Engineered a data-driven, responsive front-end using React and Next.js, integrated with a containerized Java/Spring Boot backend to deliver low-latency analytics for equity derivatives.
Leveraged AWS and Docker to build a highly available, fault-tolerant infrastructure capable of processing complex back-testing tools and large-scale financial datasets.

Technology stack:
Spring Boot, Spring MVC, Spring JPA, RESTful Web services, Eclipse/Intellij, PostgreSQL, React, Next.js, Docker, AWS

Citigroup – Cross Product Shared Services

Aug 2009- Aug 2023

System development, delivery & operational responsibility for Data Distribution Platforms, Enterprise Reference Data and Analytics Framework. These are highly distributed, high throughput, low latency, enterprise Scale solutions built using middleware, messaging architecture, caching and integration tools

Data Distribution Platform is the data distribution channel for derived market, trade and position data that is shared and consumed across various cross product areas including Credit Risk and Rates Risk Technology.
It provided high throughput, high availability infrastructure for low-latency data distribution across multiple geographical locations in North Americas, AMEA and APAC regions.

Enterprise Reference Data - Development and Integration design for the Enterprise Reference Data Services that provides reference data to various trade capturing and STP applications in Credit, Rates, FX and Derivative systems across various lion-of-businesses in Global Markets.

Responsibilities:

- Contributed to application architecture design by defining system components, data flow, and integration patterns.
- Developed services and Kafka listeners to consume notifications from external applications.
- Built highly scalable Spring Boot and Kafka Streams applications for real-time data processing.
- Designed and consumed RESTful microservices using Spring Boot to facilitate seamless data sharing and transfer between microservices.
- Modernization of technology stack from monolithic to microservices architecture.
- Migration from Tibco EMS to Kafka
- Performed debugging and root cause analysis for production issues, ensuring timely issue resolution.
- Provided technical suggestions and best practices to other team members. Conducted code reviews of other team members pull requests to maintain coding standards and quality.

Technology stack: Kafka, Tibco EMS, Gemfire, Coherence, Oracle, Sybase, Java/J2EE, JMX/MBeans, Spring

Boot, Spring Batch, Spring MVC, Spring JPA, RESTful Web services, Eclipse/Intellij, SQL, Kafka, Cucumber, Maven, Git/Github, ELK, JIRA, Jenkins, Maven

Barclays Capital (ex-Lehman Brothers)

Feb 2007- Aug 2009

Executed and delivered various initiatives around the short-term products – VRDN and TECP (Tax-exempt Commercial Paper) for short-term products front office. Delivered application systems to automate the complete trading life-cycle of Short-term products from syndication and issuance to pricing, rates dissemination, tendering, invoicing and billing.

Successfully executed and delivered the firm's highly valued STaMMP system for Auction Rate Securities, which is the firm's Auction based Trading Platform

Managed a suite of applications servicing the Municipal Short-term – MuniShop, Tender Track, CBP (Invoice and Billing). Designed the applications using Spring, Hibernate and Tibco technologies and actively contributed to the hands on development of these applications.

Technology stack: Java, C#/.Net Framework/WinForms, MyEclipse IDE, MS Visual Studio, Weblogic 8.1/9.2, Oracle 9/10, TimesTen, XmlBeans, Crystal Reports 10/XI, JMX/Mbean, JTest, Tibco EMS and RV

Prior Experience

2006– 2007: Citigroup – Operational Risk

2005 – 2006 : Lehman Brothers

Consulting projects at various banks and financial institutions - JPMorgan Chase, Salomon Smith Barney (Citi / MSSB), FirstUSA Bank, Norwest Bank

Education

Bachelor of Engineering in Computer Science from Delhi College of Engineering, New Delhi, India (1996)

Certifications & Training

- MERN Stack: MongoDB, Express.js, React, Node.js
- DevOps Tool: GitHub, BitBucket, Jenkins, Ansible, Docker, Prometheus, ELK stack
- Bond Basics and Municipal Bonds - Lehman Brothers
- Back-office trade processing and settlement - Lehman Brothers
- Developing with TIBCO Enterprise Message Services 4.x - Lehman Brothers
- Training in Equities Trading from FT Knowledge – New York Institute of Finance
- Sun Certified Business Component Developer (EJB Certification)
- BEA Weblogic Server 8.1: Covering advanced topics like clustering, monitoring, fail over and security - Citigroup
- Building Corporate Portals and Portlet Development in Java - Oracle
- Business Intelligence, Enterprise Integration and Web Services - Oracle
- Best Practices in J2EE - Oracle
- Certified as Masters in C++ by Brainbench.
- Certified as Masters in Enterprise Java Beans (EJB) by Brainbench. 4.77/5
- Brainbench Certified Web Developer in all areas, Client-Side, Server-Side and Database.

Note: No sponsorship required